Cover letter — Article submission, Scientific Reports

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Date: 2026-10-31

SUBJECT: Submission of an original Article to Scientific Reports — "Discrimination transports, calibration does not: calibration drift in severity scoring across 187 intensive-care units", by Junzhi Chu and Tiantian Zhang (24 October 2026 draft).

Dear Editors,

We are pleased to submit the above Article for consideration in Scientific Reports. Using two publicly available, de-identified intensive-care databases (eICU-CRD v2.0, 208 hospitals; MIMIC-IV v3.1, one academic centre), the study quantifies how the calibration of three standard severity scores (SOFA, OASIS, SAPS-II) varies across centres while their discrimination does not, why the drift cannot be removed by naive local recalibration on small samples, and how an empirical-Bayes shrinkage strategy recovers that loss and transfers to a second database without refitting.

The clinical stake is the assumption that institutional benchmarking by observed-to-expected ratios is meaningful. Our central result — observed-to-expected mortality spanning 0.69 to 1.54 across centres under one model, between-centre $I^2 = 75.4\%$ — directly quantifies that premise, and the bedside effect is shown by the analysis at a 0.20 decision threshold: the observed mortality among flagged patients spans 15% to 52% across centres. The study goes beyond describing the drift: the variation is partitioned into measured case-mix, centre size and calendar-year components, translated into decision-threshold consequences, and a reusable recalibration strategy is delivered with the minimum local sample size quantified.

Reporting and transparency. The manuscript reports STROBE with the RECORD extension (checklists attached), the prespecified analyses superseded from two archived, time-stamped plans (supplementary note), and four confirmatory analyses are labelled as such — including two negatively reported findings. We have kept the interpretation inside the data: the residual between-centre variation is not attributed to any difference in quality of care; the external MIMIC-IV analysis tests transferability of the shrinkage parameters rather than multicentre external validation; and the recalibration strategy is a statistical recommendation, not a clinical one.

Statements. The manuscript is not under consideration elsewhere and has not been posted to a preprint server (the analysis code is publicly archived at https://doi.org/10.5281/zenodo.23479141). All authors have read and approved the submission, declare no competing interests, and received no external funding (the article processing charge is covered by the authors' institutions). Both databases are de-identified and accessible only through PhysioNet under a Data Use Agreement; our data-use training record numbers are stated in the Ethics section of the Methods, and the authors' institutional review office confirmed that submission of this secondary analysis of published data did not require additional ethics approval. Extraction code, a centre-level summary table with no patient data (Supplementary) and all intermediate result files are provided so that every number in the manuscript can be reproduced.

Suggested reviewers, with a conflict-of-interest statement, are appended below and will be entered in the submission system. We would be glad to respond to any query or revision at the editor's discretion.

Sincerely, on behalf of all authors,

Tiantian Zhang (corresponding author), for Junzhi Chu and Tiantian Zhang

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Appendix — suggested reviewers (to be entered in the submission system)

Prof. Maarten van Smeden — University Medical Center Utrecht, the Netherlands

M.vanSmeden@umcutrecht.nl — statistician behind sample-size methods (pmsampsize) and model-validation strategy for clinical prediction models (BMC Medicine 2023).

Dr. Erik W. van Zwet — Leiden University Medical Center, the Netherlands

E.W.van_Zwet@lumc.nl — statistical foundations of shrinkage estimation (Stat Med 2021).

Prof. Mark A. van de Wiel — Amsterdam University Medical Centers, the Netherlands

mark.vdwiel@amsterdamumc.nl — shrinkage-calibration trade-off in prediction ("Think before you shrink", Stat Methods Med Res 2025).

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r. Ryohei Yamamoto — Massachusetts Institute of Technology, USA / Fukushima Medical University Hospital, Japan (ryoheiyamamoto11@gmail.com) — cross-database (MIMIC-IV versus eICU-CRD) external validation and calibration transferability (medRxiv 2026).

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rof. Gary S. Collins — University of Birmingham, UK (g.s.collins@bham.ac.uk) — professor of medical statistics and lead author of the TRIPOD+AI statement; reporting standards for prediction-model research.

Conflict-of-interest statement regarding the suggested reviewers: none of the suggested reviewers has co-authored with the authors of this manuscript, is affiliated with the authors' institutions, or has been involved in the design, analysis, or interpretation of this study.

No reviewers are excluded at this stage.